

# Mathematics for Aerospace Engineers

## Chapter 6: Intro to Partial Differential Equations and Fourier series

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## 6 Partial Differential Equations (PDEs)

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# 6 Partial Differential Equations (PDEs)

## 6.1 Introduction

PDEs are differential equations in which there is more than one independent variable. They arise in the modelling of a wide range of physical phenomena including electromagnetism, fluid dynamics, elasticity and heat conduction. The ability to solve them is of considerable importance!

In this module, we will concentrate on the use of an important technique for the solution of linear PDEs (i.e. PDEs in which the dependent variable and its derivatives appear only in a linear fashion). Important examples of linear PDEs are:

$$\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0, \quad \text{Laplace's equation,} \quad (1)$$

$$\frac{\partial^2 \phi}{\partial t^2} - c^2 \frac{\partial^2 \phi}{\partial x^2} = 0, \quad \text{the wave equation,} \quad (2)$$

$$\frac{\partial \phi}{\partial t} - k \frac{\partial^2 \phi}{\partial x^2} = 0, \quad \text{the heat or diffusion equation.} \quad (3)$$

These are also homogeneous equations, i.e. we can use the principle of superposition. Recall that if we have two solutions  $\phi_1$  and  $\phi_2$  to a linear homogeneous equation, then

$$\phi = a\phi_1 + b\phi_2$$

is also a solution for any constants  $a$  and  $b$ .

The superposition principle applies to homogeneous PDEs with linear homogeneous boundary/initial conditions. So, for example, we can apply the superposition principle to solutions satisfying the homogeneous set of boundary conditions

$$\left. \frac{\partial \phi}{\partial x} \right|_{x=0} = 0, \quad \phi|_{x=\ell} = 0.$$

## 6.1.1 Initial and boundary conditions

Before looking at solution methods for some PDEs, a brief remark on initial and boundary conditions.

Recall that when solving a second order ODE we could impose either initial conditions or boundary conditions in order to obtain a unique solution. In the case of PDEs, the choice of initial/boundary conditions for a given equation is much more complicated than for an ODE.

For example, we could specify function values on the boundary of a region, or some directional derivative on the boundary. Such boundary conditions may or may not lead to a unique solution.

Details are far beyond the scope of this module. We will just consider the cases that are most common in engineering problems.

## 6.2 Solution by separation of variables

In the case of a PDE with two independent variables,  $(x, t)$  or  $(x, y)$  say, we can *separate variables* by looking for solutions of the form

$$\phi(x, t) = g(x) f(t) \quad \text{or} \quad \phi(x, y) = g(x) f(y). \quad (4)$$

To understand why this is useful, consider the PDE

$$\frac{\partial^2 \phi}{\partial x \partial t} + \phi \sin x = 0.$$

Substituting (4) gives

$$g'(x) f'(t) + g(x) f(t) \sin x = 0.$$

Rearranging this such that all terms depending on  $t$  are on the LHS, and all terms depending on  $x$  are on the RHS,

$$\frac{f'(t)}{f(t)} = -\frac{g(x)}{g'(x)} \sin x.$$

This equation must be true for all values of  $t$  and  $x$ . This means that the LHS and RHS must be a constant. Hence,

$$\frac{f'}{f} = C \quad \text{and} \quad -\sin x \frac{g}{g'} = C,$$

where the constant  $C$  is as yet undetermined.

The separation procedure has turned the PDE into two ODEs

$$f' - C f = 0 \quad \text{and} \quad g' + \frac{\sin x}{C} g = 0.$$

The possible values for  $C$  and the complete solution will depend on both initial and boundary conditions.

## 6.3 The wave equation

Suppose we wish to find a solution to the wave equation (2) subject to the *homogeneous boundary conditions*

$$\phi(0, t) = 0, \quad \phi(L, t) = 0. \quad (5)$$

Then we can substitute (4) into (2) to obtain

$$f''(t)g(x) = c^2 g''(x)f(t) \Rightarrow \frac{f''(t)}{f(t)} = c^2 \frac{g''(x)}{g(x)}.$$

Observe that  $f''/f$  is a function of  $t$  only while  $c^2 g''/g$  is a function of  $x$  only. It follows that they must both be equal to the same constant, say  $-c^2 \lambda$ . Hence,  $f$  and  $g$  satisfy the ODEs

$$f''(t) + c^2 \lambda f(t) = 0, \quad g''(x) + \lambda g(x) = 0. \quad (6)$$



Furthermore, the boundary conditions (5) mean that we are looking for  $g$  satisfying the *eigenvalue* problem

$$g''(x) + \lambda g(x) = 0, \quad g(0) = 0, \quad g(L) = 0. \quad (7)$$

Our aim is to choose  $\lambda$  such that we obtain a *non-trivial solution* for  $g$ . The solutions to the ODE for  $g$  take the following forms, depending on the sign of  $\lambda$ :

$$g(x) = A \cos(\sqrt{\lambda}x) + B \sin(\sqrt{\lambda}x) \quad \text{for } \lambda > 0,$$

$$g(x) = A \exp(\sqrt{-\lambda}x) + B \exp(-\sqrt{-\lambda}x) \quad \text{for } \lambda < 0,$$

$$g(x) = Ax + B \quad \text{for } \lambda = 0.$$

The non-trivial solutions of (7) (i.e. the only solutions that do not give  $g(x) = 0$ ) are

$$\lambda > 0, \quad A = 0, \quad \text{and} \quad \sqrt{\lambda} = \frac{n\pi}{L}, \quad n \text{ a positive integer.}$$

Hence,

$$g(x) = B \sin \left( \frac{n\pi}{L} x \right).$$

The corresponding solution for  $f(t)$  is

$$f(t) = C_n \cos \left( \frac{n\pi c}{L} t \right) + D_n \sin \left( \frac{n\pi c}{L} t \right),$$

and the solution for the wave equation with the homogeneous boundary conditions is

$$\phi_n(x, t) = \sin \left( \frac{n\pi}{L} x \right) \left[ C_n \cos \left( \frac{n\pi c}{L} t \right) + D_n \sin \left( \frac{n\pi c}{L} t \right) \right].$$

Since the wave equation is linear and homogeneous, and has homogeneous boundary conditions, we can superpose solutions to obtain other solutions. The general solution is thus

$$\phi(x, t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi}{L}x\right) \left[ C_n \cos\left(\frac{n\pi c}{L}t\right) + D_n \sin\left(\frac{n\pi c}{L}t\right) \right].$$

The solution is still undetermined (non-unique). In order to determine it uniquely we need to prescribe initial conditions in addition to the boundary conditions.

Typically,  $\phi$  and its derivative with respect to  $t$  are given at  $t = 0$ . Call these two functions  $\tilde{\phi}$  and  $\tilde{\phi}_t$ . Then

$$\begin{aligned}\phi(x, 0) &= \tilde{\phi}(x) = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi}{L}x\right), \\ \frac{\partial}{\partial t}\phi(x, 0) &= \tilde{\phi}_t(x) = \sum_{n=1}^{\infty} \left(D_n \frac{n\pi c}{L}\right) \sin\left(\frac{n\pi}{L}x\right).\end{aligned}$$

We now have to determine the coefficients  $C_n$  and  $D_n$  from the functions  $\tilde{\phi}$  and  $\tilde{\phi}_t$ .

## Note

$C_n$  and  $D_n \left( \frac{n\pi c}{L} \right)$  are Fourier coefficients for  $\tilde{\phi}$  and  $\tilde{\phi}_t$ , respectively, and can be determined by methods discussed in Fourier analysis. We will see the initial ideas of this theory at the end of this chapter.

You may have already studied Fourier series in another module. You will also find information about them in the recommended mathematics texts for the module.

## 6.4 Laplace's equation

In general, Laplace's equation is

$$\nabla^2 f = 0.$$

If  $f$  is a function of two variables  $x$  and  $y$ , then

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0.$$

Using separation of variables, let  $f(x, y) = h(x)g(y)$  and substitute this into Laplace's equation,

$$h''g + hg'' = 0 \quad \Rightarrow \quad \frac{h''(x)}{h(x)} = -\frac{g''(y)}{g(y)} = C,$$

where  $C$  is a constant. Hence

$$h'' - Ch = 0 \quad \text{and} \quad g'' + Cg = 0.$$

If  $C > 0$ , i.e.  $C = a^2$  with a real constant  $a$ , then the solutions are

$$h(x) = k_1 e^{ax} + k_2 e^{-ax}, \text{ and } g(y) = k_3 \cos(ay) + k_4 \sin(ay).$$

If  $C < 0$ , i.e.  $C = -a^2$ , then the solutions are

$$h(x) = k_1 \cos(ax) + k_2 \sin(ax), \text{ and } g(y) = k_3 e^{ay} + k_4 e^{-ay}.$$

If  $C = 0$ , then the solutions are

$$h(x) = k_1 x + k_2, \text{ and } g(y) = k_3 y + k_4.$$

In general, the solution will be a superposition of such functions with different values of  $a$ . Which values of  $a$  appear in the solution depends on the boundary conditions.

Examples of functions  $f$  that satisfy Laplace's equation are

$$f(x, y) = e^{\pm ax} \sin ay,$$

$$f(x, y) = e^{\pm ax} \cos ay,$$

$$f(x, y) = e^{\pm ay} \sin ax,$$

$$f(x, y) = e^{\pm ay} \cos ax.$$

(see problem sheet for some solid examples).

## The diffusion equation (or heat equation)

This equation describes the time-dependent flow of heat in a material. It also describes other diffusion processes, such as the diffusion of one chemical substance in another. If the temperature (or chemical concentration) is  $T(x, y, z, t)$ , the diffusion equation is

$$\frac{\partial T}{\partial t} = k \nabla^2 T,$$

where  $k > 0$  is a constant called the diffusivity.

If there is only one spacial dimension (e.g. the flow of heat along a bar), the equation becomes

$$\frac{\partial T}{\partial t} = k \frac{\partial^2 T}{\partial x^2}.$$



Time to separate variables! Let  $T(x, t) = f(t)g(x)$  and substitute into the heat equation,

$$f'g = kfg'', \quad \Rightarrow \quad \frac{f'}{kf} = \frac{g''}{g} = C,$$

where  $C$  is a constant. Hence,

$$f' - Ckf = 0 \quad \text{and} \quad g'' - Cg = 0.$$

The solution for  $f$  is  $f(t) = Ae^{Ckt}$ , but we note  $f(t) \rightarrow \infty$  as  $t \rightarrow \infty$  (which is not physical) if  $C > 0$ , and  $f(t) = \text{const}$  if  $C = 0$  (which is boring), so we take  $C = -a^2$  with a real constant  $a$ . Then

$$f(t) = Ae^{-ka^2t} \quad \text{and} \quad g'' + a^2g = 0,$$

with solutions

$$g(x) = B\cos(ax) + D\sin(ax).$$

The general solution is

$$T(x, t) = e^{-ka^2t}[\alpha \cos(ax) + \beta \sin(ax)],$$

where the constants  $\alpha$ ,  $\beta$  and  $a$  are determined by initial and boundary conditions.

An example of a solution is

$$T(x, t) = e^{-ka^2t} \sin ax,$$

for any constant  $a$ .

See problem sheet for some more examples.

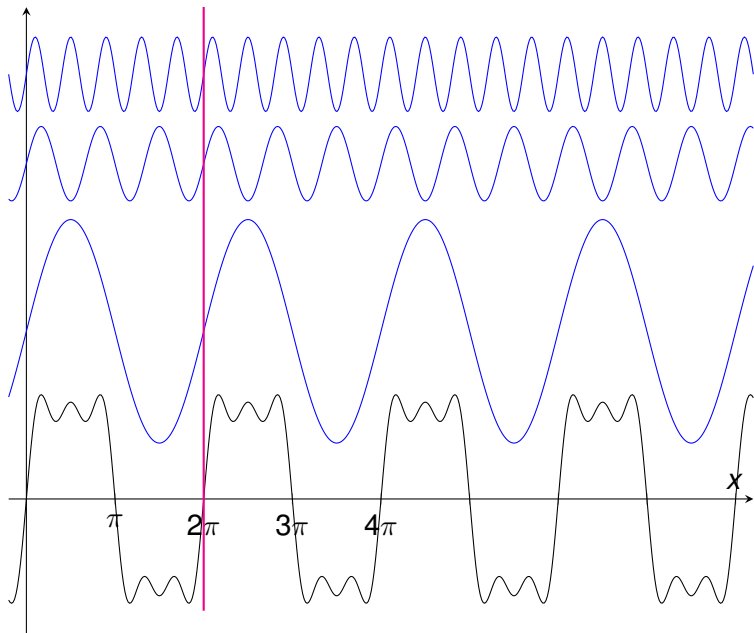
## 6.6 Introduction to Fourier series

The idea of a **Fourier series** is to express a periodically repeating function as a sum of sine and cosine waves (and maybe a constant term too).

We all know that sines and cosines periodically repeat (precise definition of 'periodicity' later). If you add two or more with 'commensurate' frequency, then the sum still periodically repeats.

The idea is that we can then build up more complicated periodically repeating functions using (usually infinite) sums of the more familiar sines and cosines.

On the next slide we have plotted the graphs of  $\sin(x)$ ,  $\sin(3x)/3$  and  $\sin(5x)/5$  (in blue) as well as their sum (in black). The blue curves have been shifted upwards to avoid overlaps.

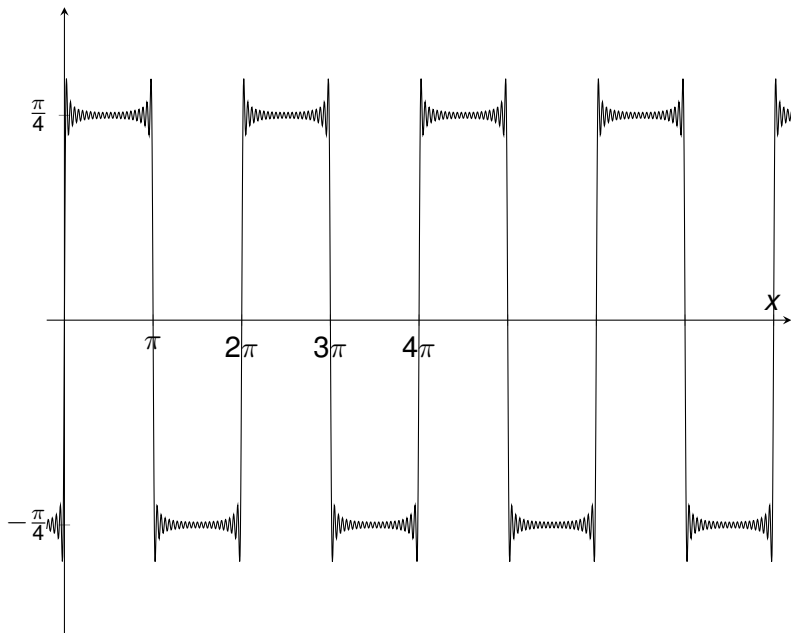


Some things to notice:

- 1 All curves repeat every  $2\pi$  units. In other words, the section between 0 and  $2\pi$  (or any interval of length  $2\pi$ ) can be copied and pasted across the graph to recover everything.
- 2 Although the higher frequency sine waves repeat themselves even sooner, it is still true that they repeat themselves every  $2\pi$  units (but also with shorter periods for higher frequencies).
- 3 The higher frequencies (faster oscillations) are chosen here with smaller amplitudes. In Fourier Analysis, higher frequency oscillations become small for reasonable functions (so the main contribution comes from the lower frequencies).

We can consider adding more and more sine or cosine waves together. On the next slide we continue the above pattern and plot

$$\sin(x) + \frac{\sin(3x)}{3} + \frac{\sin(5x)}{5} + \frac{\sin(7x)}{7} + \dots + \frac{\sin(43x)}{43}.$$



As we add more terms the graph is looking closer to the 'square wave function' which jumps periodically between  $\frac{\pi}{4}$  and  $-\frac{\pi}{4}$ . Indeed, in the infinite limit, this square wave can literally be written:

$$f(x) = \sum_{n=1}^{\infty} \frac{\sin((2n-1)x)}{2n-1} = \sin(x) + \frac{\sin(3x)}{3} + \frac{\sin(5x)}{5} + \dots$$

This is an example of a **Fourier series**.

We can get a reasonable approximation by 'truncating' the sum to a finite one (higher frequency oscillations have smaller effect). That is, we may consider the **Fourier sum**

$$f_N = \sum_{n=1}^N \frac{\sin((2n-1)x)}{2n-1} = \sin(x) + \frac{\sin(3x)}{3} + \frac{\sin(5x)}{5} + \dots + \frac{\sin((2N-1)x)}{2N-1},$$

(the graph on the previous slide was of  $f_{22}$  since  $2 \cdot 22 - 1 = 43$ ).

This is a similar concept to Taylor series, except we approximate with sines and cosines rather than polynomials!

Some further observations on the last graph:

- 1 Even though our infinite sum involved nice and smooth sine waves, the limiting infinite sum has 'jump discontinuities', which is very surprising!
- 2 What does the infinite sum do exactly at the points of discontinuity? It turns out it decides to use the midpoint of the left and right limits i.e., it evaluates to 0 at integer multiples of  $\pi$ .
- 3 You may notice the 'bunny ears' of high amplitude fine oscillations near to the jump discontinuities. These never go away, but are pushed to smaller and smaller regions around the jump discontinuities as we add more and more sine terms. They only disappear in the infinite limit. This is called a **Gibbs Phenomenon**.

In the following slides we systemise the ideas seen above for this particular example of a square wave, starting with the idea of a 'periodic function'.



## 6.7 Periodic functions

A function  $f$  of one variable  $x$  is said to be **periodic** if there exists a non-zero fixed number  $p$  (called the **period**) such that

$$f(x) = f(x + p) \quad \text{for all } x.$$

For example,  $\sin(\frac{\pi}{\ell}x)$  and  $\cos(\frac{\pi}{\ell}x)$  are  $2\ell$ -periodic.

In Engineering, such functions are extremely important: Oscillations of all kinds are described by periodic functions.

Periodic functions describing oscillations will often be far more complicated than the sine and cosine functions describing simple harmonic oscillations, and they may often be hard to deal with in calculations, eg if they turn up in a differential equation, describing the force driving some vibration or an initial condition for a PDE.

**Note:**  $\ell$  will always denote **half** of the period. This will be a useful convention!

## 6.8 Fourier series

An infinite sum (or finite sum, if only finitely many  $a_n$  and  $b_n \neq 0$ ) of the form

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left( a_n \cos\left(\frac{n\pi}{\ell}x\right) + b_n \sin\left(\frac{n\pi}{\ell}x\right) \right)$$

is called a **Fourier series** with period  $2\ell$ .

The constants  $a_n$  and  $b_n$  are called the **Fourier coefficients**.

Note that the above infinite sum will not converge to a limit for most values of  $x$  if the Fourier coefficients do not shrink to 0 as  $n \rightarrow \infty$ .

When they do and  $f(x)$  is defined for all  $x$ , it is not too hard to see that  $f(x)$  is  $2\ell$ -periodic.

In short: a Fourier series is just a weighted sum (maybe infinite) of a constant, plus cosine and sine waves that oscillate exactly  $n$  times across one period, for positive integers  $n$ .

Given a  $2\ell$  periodic function  $f(x)$ , we would like to know how to express it as a Fourier series i.e., to find the Fourier coefficients. They may be computed by the following integrals:

### Fourier series for a $2\ell$ -periodic function $f$

$$\text{For } f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \left( a_n \cos \left( \frac{n\pi}{\ell} x \right) + b_n \sin \left( \frac{n\pi}{\ell} x \right) \right),$$

$$a_0 = \frac{1}{\ell} \int_{-\ell}^{\ell} f(x) \, dx$$

$$a_n = \frac{1}{\ell} \int_{-\ell}^{\ell} f(x) \cos \left( \frac{n\pi}{\ell} x \right) \, dx \quad (n \geq 1)$$

$$b_n = \frac{1}{\ell} \int_{-\ell}^{\ell} f(x) \sin \left( \frac{n\pi}{\ell} x \right) \, dx \quad (n \geq 1)$$

I will outline a proof of this in lectures. It is a good idea to memorise the formulae for the Fourier coefficients (there are some simple patterns in these formulae).

## Example

Obtain the Fourier series for the  $2\pi$ -periodic triangle wave

$$f(x) = \begin{cases} x + \frac{\pi}{2} & \text{for } -\pi < x < 0, \\ \frac{\pi}{2} - x & \text{for } 0 \leq x \leq \pi. \end{cases}$$

In this example,  $\ell = \pi$ .

All we need to do is to substitute  $f(x)$  into the equations for  $a_0$ ,  $a_m$  and  $b_m$  on the previous slide!

$$a_0 = 0,$$

$$a_m = \frac{1}{\pi} \left( \int_{-\pi}^{\pi} \frac{\pi}{2} \cos(mx) \, dx + \int_{-\pi}^0 x \cos(mx) \, dx + \int_0^{\pi} (-x) \cos(mx) \, dx \right) \quad \text{for } m \geq 1,$$

$$b_m = \frac{1}{\pi} \left( \int_{-\pi}^{\pi} \frac{\pi}{2} \sin(mx) \, dx + \int_{-\pi}^0 x \sin(mx) \, dx + \int_0^{\pi} (-x) \sin(mx) \, dx \right) = 0.$$

Note that the first integrals in the expressions for both  $a_m$  and  $b_m$  are zero and that the second and third integrals in the expression for  $a_m$  take the same value whilst those in the expression for  $b_m$  cancel. Thus

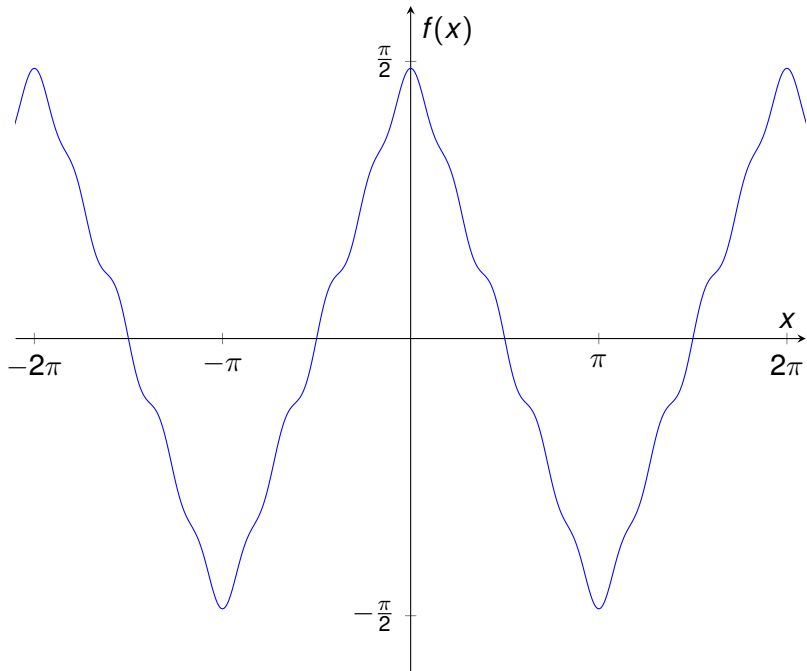
$$a_m = -\frac{2}{\pi} \int_0^{\pi} x \cos(mx) \, dx = -\frac{2}{\pi} \left[ \frac{x \sin(mx)}{m} + \frac{\cos(mx)}{m^2} \right]_0^{\pi}$$

$$\Rightarrow a_m = \frac{2}{m^2 \pi} (1 - (-1)^m),$$

and the Fourier series for  $f(x)$  is

$$f(x) = \sum_{m=1}^{\infty} \frac{2}{m^2 \pi} (1 - (-1)^m) \cos(mx).$$

A plot of the first 10 terms in the Fourier series is shown in the following graph:



## Remark

Note that  $f$  here is an **even function** ( $f(-x) = f(x)$  i.e., looks the same reflected over the  $y$ -axis) and its Fourier series contains no  $\sin(mx)$  terms.

It is in general true that the Fourier series for an even function  $f$  has all  $b_n = 0$  so takes the form

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{\ell}x\right),$$

while an **odd function**  $g$  (that is,  $g(-x) = -g(x)$  i.e., it looks the same after rotating 180 degrees around the origin) has all  $a_n = 0$  so takes the form

$$g(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{\ell}x\right),$$

(convince yourself that these statements are true). Remembering these facts can save time when evaluating the Fourier coefficients.



# Fourier's Theorem

Can we rely on the Fourier series actually approximating our original periodic function? **Fourier's Theorem** essentially says you can:

## Fourier's Theorem

For any reasonable\*  $2\ell$ -periodic function  $f$ :

- at points  $x = a$  where  $f$  is continuous (no 'jump' in the graph), the sequence of Fourier sums tend to  $f(a)$ ;
- at points  $x = b$  where  $f$  is discontinuous (the graph jumps from value  $f(b^-)$  on the left to  $f(b^+)$  on the right), the sequence of Fourier sums tend to the midpoint

$$\frac{1}{2}(f(b^+) + f(b^-)).$$

\* EXTRACURRICULAR SMALL-PRINT: I do not expect you to know exactly what 'reasonable' means, but it will hold for any functions you should encounter in applications. Even continuous functions (those with no 'jumps' in the graph i.e., those whose graphs you can draw without lifting your pen) can have such wild oscillations so as to not have Fourier sums correctly converging everywhere. But such functions need to oscillate wildly and won't show up in applications.

## Fourier series of functions on finite intervals

Are Fourier series only useful for periodic functions? No! In many applications, you will only care about the value of a function  $f(x)$  on a finite interval, say  $0 < x < \ell$ .

Since the function is initially undefined outside this interval we can

- (i) choose it to be periodic with period  $2\ell$  i.e.  $f(x + 2\ell) = f(x)$ , and
- (ii) extend the definition into the interval  $-\ell < x < 0$  by choosing

$$\begin{aligned} & f \text{ to be even } (f(-x) = f(x)) \\ \text{or} \quad & f \text{ to be odd } (f(-x) = -f(x)). \end{aligned}$$

Note that we do not have to specify  $f(0)$ , as it has no influence on the Fourier coefficients: If we choose the even extension, the Fourier series will converge to  $\lim_{x \rightarrow 0} f(x)$  resulting in a function that is continuous at  $x = 0$ . If we choose the odd extension, the Fourier series will converge to zero at  $x = 0$ . Can you see why?

Choosing  $f$  to be even in  $x$  implies that  $f$  can be represented by a Fourier cosine series so that

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{\ell}\right),$$

$$a_n = \frac{1}{\ell} \int_{-\ell}^{\ell} f(x) \cos\left(\frac{n\pi x}{\ell}\right) dx = \frac{2}{\ell} \int_0^{\ell} f(x) \cos\left(\frac{n\pi x}{\ell}\right) dx.$$

Alternatively choosing  $f$  to be odd in  $x$  leads to a Fourier sine series representation

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{\ell}\right),$$

$$b_n = \frac{1}{\ell} \int_{-\ell}^{\ell} f(x) \sin\left(\frac{n\pi x}{\ell}\right) dx = \frac{2}{\ell} \int_0^{\ell} f(x) \sin\left(\frac{n\pi x}{\ell}\right) dx.$$

Note the simplification of the integrals: We can do this as the integrands are even functions of  $x$ .

# Applications of Fourier series

We have already seen an application: supposing that  $\phi(0, t) = \phi(\ell, t) = 0$  for all  $t$ , the Wave Equation

$$\frac{\partial^2 \phi}{\partial t^2} - c^2 \frac{\partial^2 \phi}{\partial x^2} = 0$$

has solution:

$$\phi(x, t) = \sum_{n=1}^{\infty} \sin\left(\frac{n\pi}{\ell}x\right) \left[ C_n \cos\left(\frac{n\pi c}{\ell}t\right) + D_n \sin\left(\frac{n\pi c}{\ell}t\right) \right] \quad (8)$$

for constants  $C_n$  and  $D_n$ . Substituting  $t = 0$ :

$$\phi(x, 0) = \tilde{\phi}(x) = \sum_{n=1}^{\infty} C_n \sin\left(\frac{n\pi}{\ell}x\right).$$

Differentiating  $\phi$  term-by-term and then substituting  $t = 0$ ,

$$\frac{\partial}{\partial t} \phi(x, 0) = \tilde{\phi}_t(x) = \sum_{n=1}^{\infty} \left( D_n \frac{n\pi c}{L} \right) \sin\left(\frac{n\pi}{\ell}x\right).$$

The constants  $C_n$  are thus just the coefficients  $b_n$  of the Fourier sine series for the initial shape of the wave (at points  $0 < x < \ell$ ).

Similarly, the constants  $D_n \frac{n\pi c}{L}$  are just the coefficients  $b_n$  for the Fourier sine series of the initial velocity of the wave (at points  $0 < x < \ell$ ).

**Simple Example:** Let our wave be defined between  $0 < x < 1$ , initially at rest and with position

$$\tilde{\phi}(x) = \sin(\pi x) - 3 \sin(4\pi x).$$

What does it look like at later times?

This is a simple example, as we don't need to work out the Fourier sine series, it's already given to us! (and is finite):  $b_1 = 1$ ,  $b_4 = -3$  and  $b_n = 0$  otherwise, with the half-period  $\ell = 1$ . Then  $C_1 = 1$  and  $C_4 = -3$  in Equation 8, whilst every other  $C_n = 0$ .

Since it is initially at rest,  $D_n = 0$  for all  $n$ . Thus, the solution is:

$$\phi(x, t) = \sin(\pi x) \cos(\pi ct) - 3 \sin(\pi x) \cos(\pi ct)$$

(note that infinitely many of the coefficients are 0, so the infinite sum from Equation 8 becomes a finite sum).

Click the link below to see an animation of this wave (where the wave constant  $c$  can be varied):

<https://www.desmos.com/calculator/nfpykgt1tp>

## A more complicated example

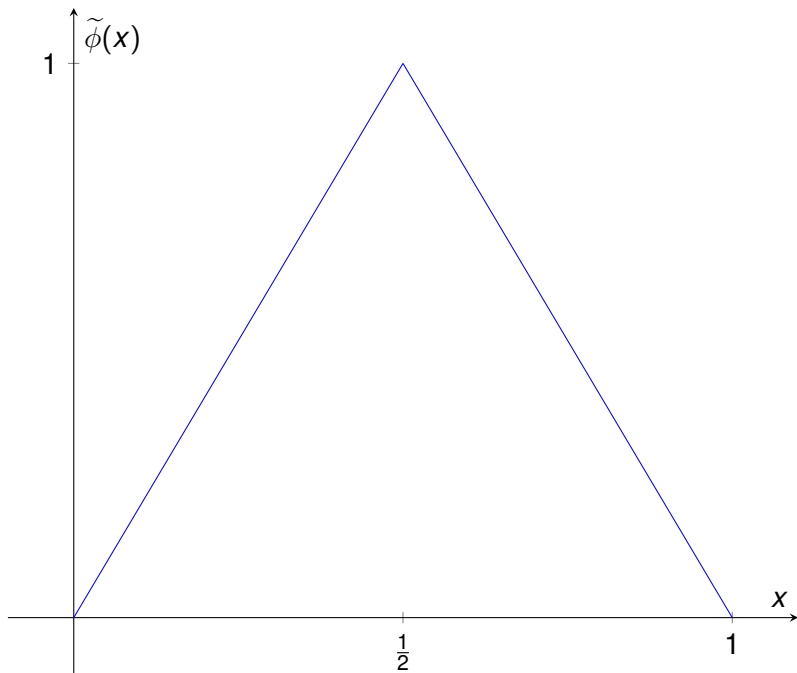
The last example did not need the machinery of Fourier series. This one will do.

Suppose our wave is defined for  $0 < x < 1$ , is initially at rest and with position

$$\tilde{\phi}(x) = \begin{cases} 2x & \text{for } 0 < x < \frac{1}{2}; \\ 2 - 2x & \frac{1}{2} < x < 1. \end{cases}$$

This is a triangular wave (see next slide). What does it look like at later times?

First we will find the Fourier sine series expansion for  $\tilde{\phi}(x)$ .





By using the odd extension of this to a 2-periodic function,

$$\tilde{\phi}(x) = \sum_{n=1}^{\infty} b_n \sin(n\pi x), \text{ where } b_n = 2 \int_0^1 \tilde{\phi}(x) \sin(n\pi x) dx$$

(see Section 9). Then:

$$\begin{aligned} b_n &= 2 \int_0^{1/2} 2x \sin(n\pi x) dx + 2 \int_{1/2}^1 (2 - 2x) \sin(n\pi x) dx = \\ &= \frac{4 \sin\left(\frac{n\pi}{2}\right) - 2n\pi \cos\left(\frac{n\pi}{2}\right)}{\pi^2 n^2} + \frac{4 \sin\left(\frac{n\pi}{2}\right) - 4 \sin(n\pi) + 2n\pi \cos\left(\frac{n\pi}{2}\right)}{n^2 \pi^2} = \\ &= \frac{8 \sin\left(\frac{n\pi}{2}\right) - 4 \sin(n\pi)}{n^2 \pi^2} = \\ &= \frac{8 \sin\left(\frac{n\pi}{2}\right)}{n^2 \pi^2}. \end{aligned}$$

In the above, we use integration by parts (details omitted), and  $\sin(n\pi) = 0$  for integers  $n$ .

Similarly,  $\sin\left(\frac{n\pi}{2}\right) = 0$  for *even* integers  $n$ . For  $n = 2k - 1$  odd,  $\sin\left(\frac{(2k-1)\pi}{2}\right) = (-1)^{k+1}$  (sine alternates between its max/min value at  $\pi/2, 3\pi/2, 5\pi/2, \dots$ ). Thus:

$$b_n = \begin{cases} 0 & \text{for } n \text{ even;} \\ \frac{8(-1)^{k+1}}{(2k-1)^2\pi^2} & \text{for } n = 2k - 1 \text{ odd.} \end{cases}$$

This allows us to write down the Fourier sine series for  $\tilde{\phi}$ . We have  $C_n = b_n$  in Eq 8, and each  $D_n = 0$  (the wave is initially at rest). Thus:

$$\phi(x, t) = \sum_{k=1}^{\infty} \frac{8(-1)^{k+1}}{(2k-1)^2\pi^2} \sin((2k-1)\pi x) \cos\left(\frac{(2k-1)\pi c}{\ell} t\right).$$

Press the below link to see an animation, with wave constant  $c = 1$ . To do this, we approximate the above by only summing  $N$  terms.

<https://www.desmos.com/calculator/cytujob8ivj>

# Application of Fourier series to ODEs with periodic RHS

We saw in Chapter 1 how to solve ODEs of the form

$$a \frac{d^2 y}{dx^2} + b \frac{dy}{dx} + cy = g(x)$$

if  $g(x)$  is of a limited form (or maybe involving Heaviside step functions or Dirac Deltas, using techniques from Chapter 2).

Fourier series will allow us to solve such ODEs if the RHS (the ‘forcing function’) is periodic or if we only need the answer on a finite interval:

- 1 Find the complementary function as usual;
- 2 Express  $g(x)$  as a Fourier series;
- 3 ‘Guess’ a particular integral in the form of a Fourier series (with the same period).
- 4 Substitute into the LHS of the ODE and find the Fourier coefficients of the particular solution by comparing coefficients.

## Example

A weighted spring with natural frequency  $\frac{\omega}{2\pi}$  is forced by the  $2\pi$  periodic forcing function

$$F(t) = \begin{cases} (t + \pi) & \text{for } -\pi \leq t < 0 \\ (\pi - t) & \text{for } 0 \leq t < \pi. \end{cases}$$

The vertical displacement  $y$  of the weight satisfies the equation

$$\frac{d^2y}{dt^2} + \omega^2 y = F(t). \quad (9)$$

Find  $y$  as a function of  $t$ .

We start by expressing  $F(t)$  as a Fourier series. Since  $F$  is an even function, we have  $b_m = 0$  for all  $m$ , and the coefficients  $a_m$  are given by

$$a_m = \frac{2}{\pi} \int_0^{\pi} (\pi - t) \cos(mt) \, dt,$$

$$\text{ie } a_0 = \pi$$

$$\begin{aligned} \text{and } a_m &= \frac{2}{\pi} \left[ \frac{\pi}{m} \sin(mt) - \frac{1}{m^2} \cos(mt) - \frac{t}{m} \sin(mt) \right]_0^{\pi} \\ &= \frac{2}{m^2 \pi} (1 - (-1)^m), \quad m \geq 1. \end{aligned}$$

Substituting the Fourier series for  $F(t)$  into (9) gives

$$\frac{d^2 y}{dt^2} + \omega^2 y = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{\pi n^2} (1 - (-1)^n) \cos(nt).$$

Now write the particular integral as

$$y_p(t) = \frac{A_0}{2} + \sum_{n=1}^{\infty} (A_n \cos(nt) + B_n \sin(nt)) ,$$

so, differentiating twice term-by-term,

$$\frac{d^2 y_p}{dt^2} = \sum_{n=1}^{\infty} \left( -n^2 A_n \cos(nt) - n^2 B_n \sin(nt) \right) .$$

Inserting all this into the ODE and grouping terms, we get

$$\begin{aligned} \frac{\omega^2 A_0}{2} + \sum_{n=1}^{\infty} \left( (\omega^2 - n^2) A_n \cos(nt) + (\omega^2 - n^2) B_n \sin(nt) \right) \\ = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{2}{\pi n^2} (1 - (-1)^n) \cos(nt). \end{aligned}$$

We can now determine  $A_n, B_n$  by comparing coefficients,

$$\left. \begin{aligned} A_0 &= \frac{\pi}{\omega^2}, \\ A_n &= \frac{2}{\pi n^2(\omega^2 - n^2)}(1 - (-1)^n) \\ B_n &= 0 \end{aligned} \right\} \quad (n \geq 1).$$

Two independent solutions to the homogeneous version of (9) (i.e. with  $F = 0$ ) are  $\cos(\omega t)$  and  $\sin(\omega t)$  and it follows that the general solution to (9) is

$$\begin{aligned} y(t) = & C_1 \cos(\omega t) + C_2 \sin(\omega t) + \frac{\pi}{2\omega^2} \\ & + \sum_{n=1}^{\infty} \frac{2}{\pi n^2(\omega^2 - n^2)}(1 - (-1)^n) \cos(nt). \end{aligned} \quad (10)$$

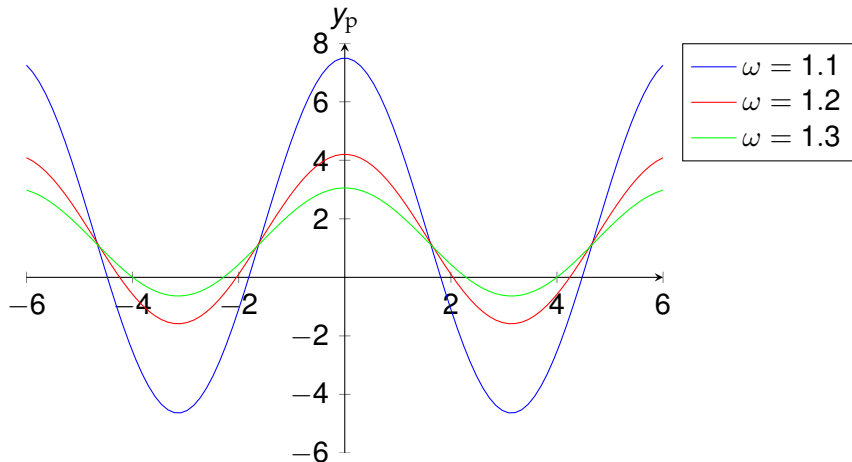


Figure 1: This picture shows  $y_p$  for  $\omega = 1.1$ ,  $\omega = 1.2$  and  $\omega = 1.3$ . Notice the increasing amplitudes as  $\omega$  approaches 1. This effect is called *resonance*.



In some more detail, resonance occurs for a given  $\omega$  if one of the Fourier coefficients gets arbitrarily large as we approach that value of  $\omega$ .

In the above example, each  $B_n = 0$  and

$$A_n = \frac{2}{\pi n^2(\omega^2 - n^2)}(1 - (-1)^n).$$

So  $A_n = 0$  for any *even*  $n$ . For *odd*  $n$ , the value of  $A_n$  gets arbitrarily large as  $\omega$  approaches  $n$ , since then the  $\omega^2 - n^2$  term of the denominator approaches 0.

**So this system has resonance for  $\omega$  an odd integer!**

See the following graph (where the number of terms in the Fourier sum and value  $\omega$  (denoted instead  $w$ ) can be varied:

<https://www.desmos.com/calculator/rntftk4oer>